

Accident Hotspot Identification Dashboard/Smart Accident Predict

Team No. 15

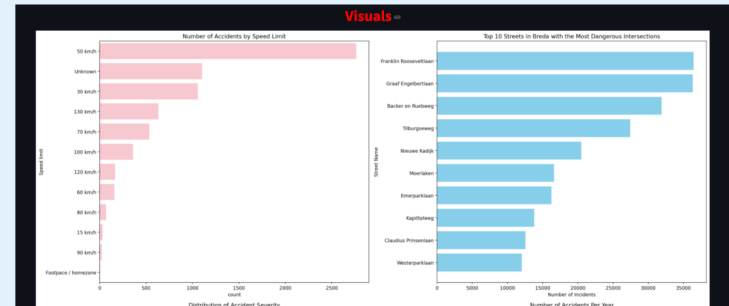
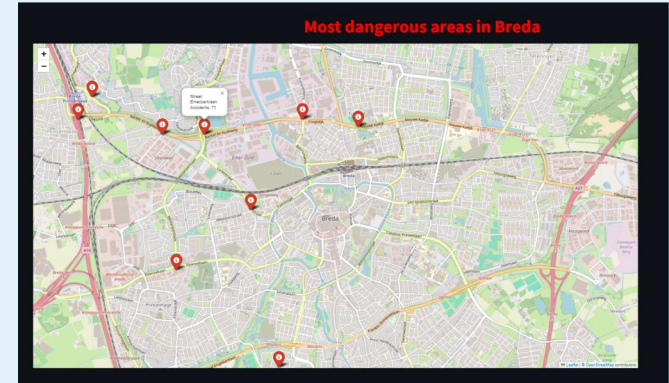
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Introduction

Project Overview

- **Purpose:** Develop an advanced dashboard integrating a machine learning model to predict the severity of traffic accidents based on specific accident details. Identify dangerous road intersections using unsupervised clustering.
- **Method:** Analyze real-time data from cameras and sensors, use clustering techniques to identify and analyze the most dangerous intersections in Breda, and predict accident severity using machine learning and deep learning models.
- **Objective:** Enhance road safety in Breda by providing actionable insights, enabling efficient mobilization of law enforcement and emergency services, and informing city planners and traffic management authorities.



Business Value Overview

Proactive Safety Measures: Predicts accident severity, allowing timely interventions to reduce accidents and improve public safety.

Efficient Resource Use: Optimizes deployment of law enforcement and emergency services to high-risk areas, enhancing operational efficiency.

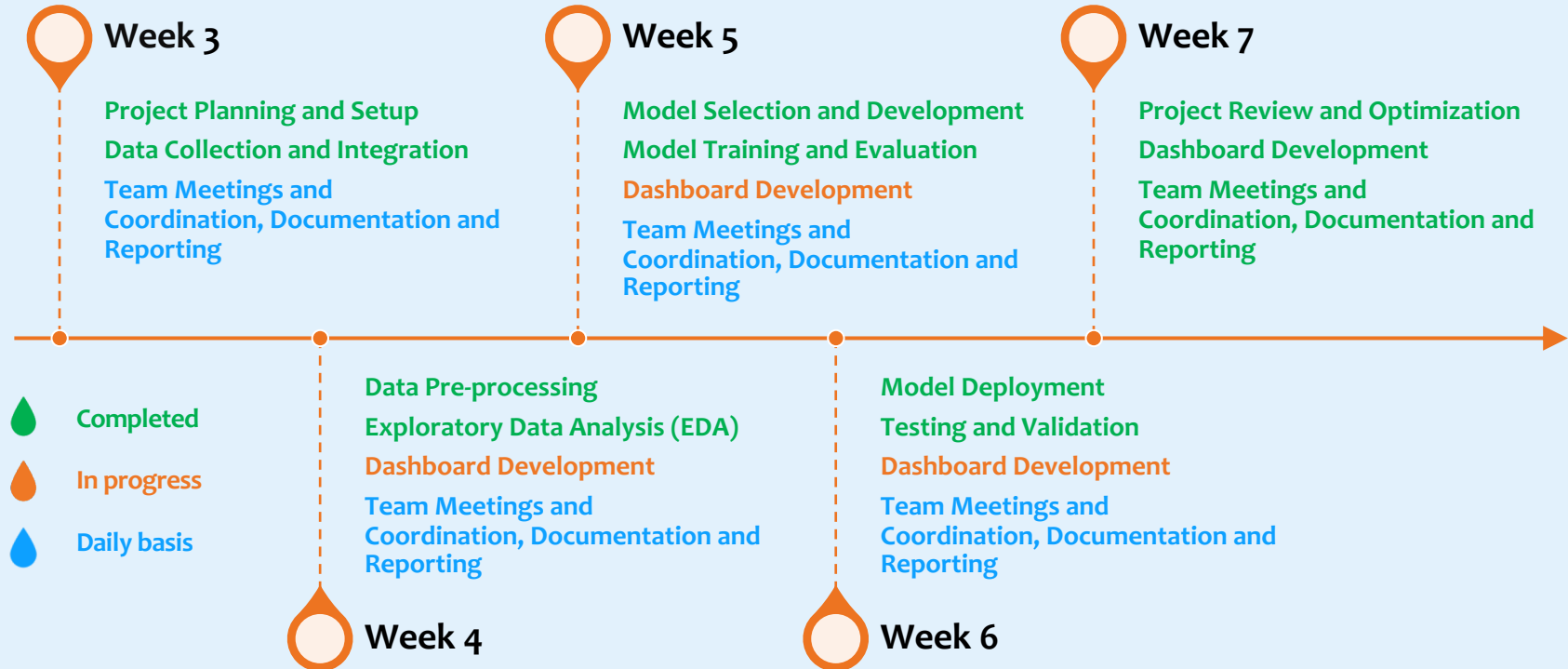
Cost Savings: Lowers healthcare, vehicle repair, and insurance costs by reducing accident frequency and severity.

Data-Driven Decisions: Informs infrastructure improvements and safety measures through actionable insights on high-risk intersections.

Community Well-Being: Enhances quality of life with fewer accidents, injuries, and fatalities.

Technological Innovation: Showcases Breda as a leader in using predictive analytics and machine learning for urban safety.

Project Timeline



Problem Statement

Problem Definition

- **Objective:** Create a model to predict and reduce traffic accident severity in Breda using machine learning.
- **Challenges:** Late arrival of emergency services risks lives and lowers service quality.
- **Approach:** Use real-time data from cameras and sensors to classify accident severity by factors like road details, date, weather, and vehicle characteristics.

Importance of the solution

- **Safety Improvement:** Reduce accident rates and enhance road safety.
- **Efficient Resources:** Better deployment of law enforcement and emergency services.
- **Targeted Measures:** Implement speed cameras and improve high-risk intersections.
- **Public Safety:** Enhance safety for all, leading to cost savings and better quality of life.

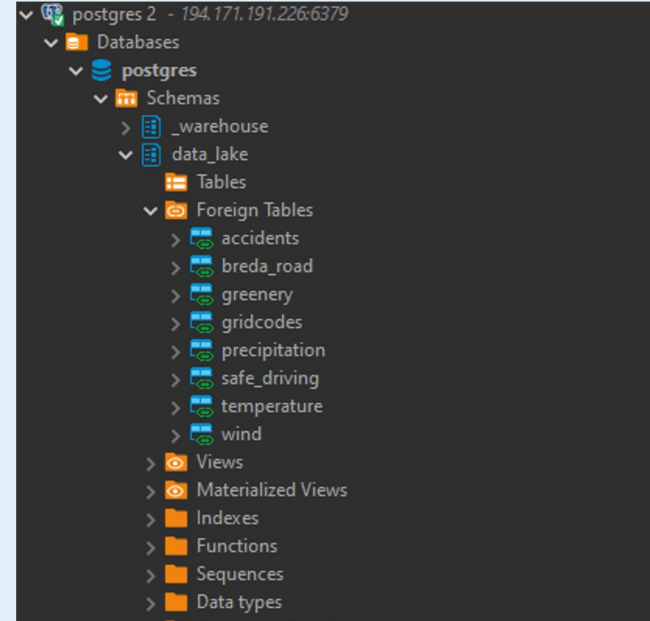
Context and Background

- **Road Safety:** Traffic accidents are a major risk despite ANWB's efforts.
- **Breda's Scenario:** Busy urban area with traffic management challenges.
- **Current Methods:** Enhance accident response with predictive analytics.

Data Description

Datasets

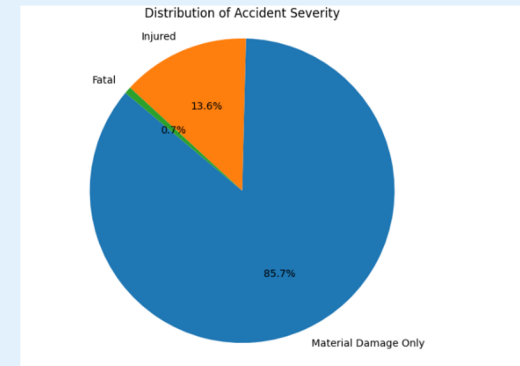
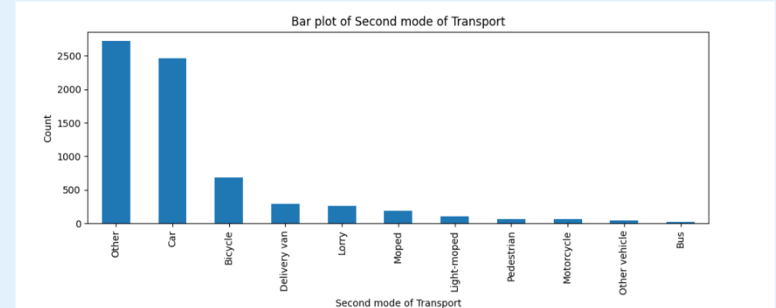
- ANWB (Safe Driving) Data – records of driving incidents (speeding, harsh cornering, hard acceleration, and abrupt braking)
- Accidents_on_each_street - preprocessed safe driving table with number of accidents on each street in Breda
- BRON (Accidents) Data - details of traffic accidents (date, time, location, severity, and conditions at the time of the accident)
- "Accidents_17_23" dataset with comprehensive information on accidents and their severity.
- Speeding - dataset with top 10 average speeding streets



Data Preparation

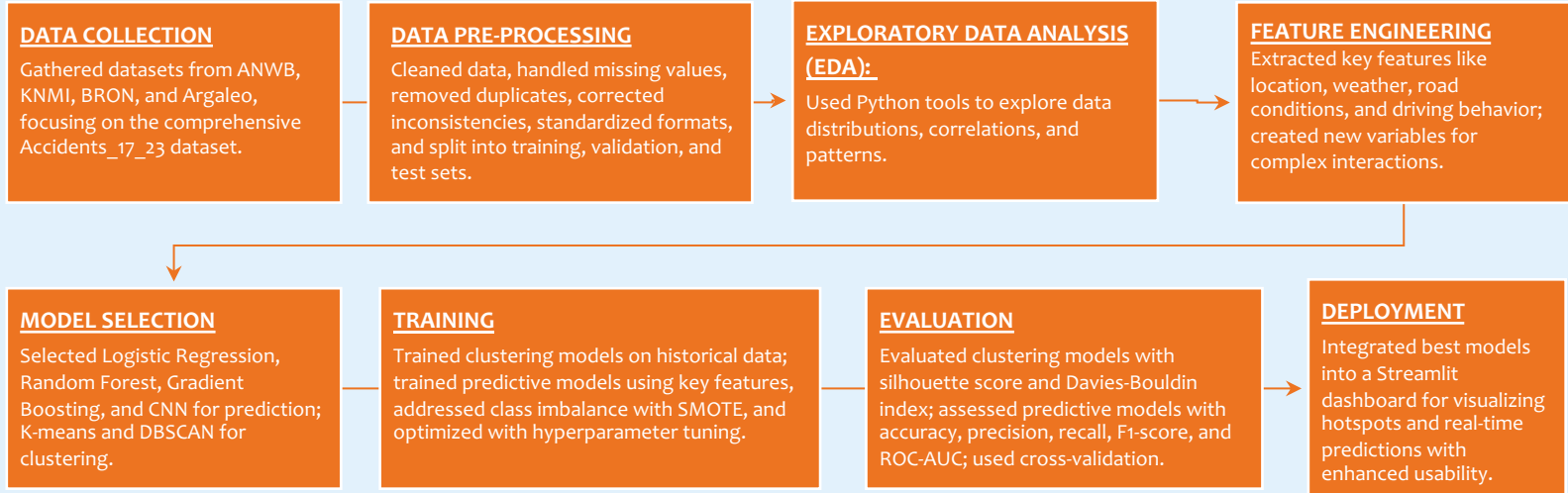
Preprocessing steps:

- Dealing with empty rows and columns
- Selecting relevant columns
- Data normalization
- Label encoding
- Feature engineering
- Merging datasets



Methodology

Steps



Machine Learning

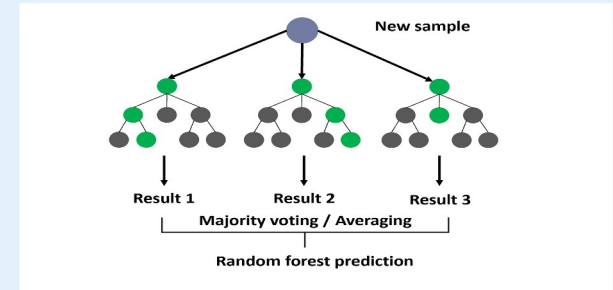
Models and Their Architecture

Clustering Model:

- **Algorithm:** K-means Clustering and DBSCAN
- **Reason:** K-means and DBSCAN are effective for spatial clustering, with K-means identifying distinct groups and DBSCAN detecting clusters of varying shapes and densities
- **Architecture:**
 - **Input Features:** Latitude, longitude, time of day, and weather conditions.
 - **Output:** Clusters representing accident hotspots.

Predictive Model:

- **Algorithm:** Logistic Regression, Random Forest, Gradient Boosting, and Convolutional Neural Network (CNN)
- **Reason:**
 - **Random Forest and Gradient Boosting:** Robust for handling complex, non-linear relationships, providing high accuracy and interpretability for classification tasks.
 - **CNN:** Effective for capturing spatial patterns and geographical distribution of accidents.
- **Architecture:**
 - **Input Features:** Historical accident data, weather data, road characteristics, and driving behaviour metrics.
 - **Output:** Probability of an accident occurring in each identified hotspot.



Severity

Back

Initial transport Car	Secondary transport Car	Area type Urban area	Light condition Daylight	Road location Road section	Road condition Dry
Road surface Asphalt (other)	Road situation Straight road	Weather Dry	Speed limit 15 km/h	Street Boulevard	<button>Predict</button>

Predicted severity: material damage only

Risks of the Project

Risk Associated with the AI System



Data Quality: Inaccurate or incomplete data could lead to poor model performance.



Model Accuracy: Incorrect predictions could result in inefficient resource allocation and potential safety risks.



Bias: The models might inadvertently learn and propagate biases present in the historical data.

Thank you for your attention!

